

RESEARCH RESPONSE

Cognitive Friction and Designing for Control in Autonomous Systems

Brandon Fluegel, PhD

Human Factors Researcher

www.humanfactors.pro

EXECUTIVE SUMMARY

This brief research strategy addresses three critical human-machine teaming questions potentially facing the deployment of autonomous systems at Anduril. Prompted by questions raised by Dr. Daniella Kim at the Learners Conference (San Francisco, May '26), this doc provides a Human Factors POV on Meaningful Human Control (MHC) and the ethical implications of User Experience (UX) design in modern warfare. Standard consumer UX practices design for seamless, frictionless interactions. However, in the context of autonomous (potentially lethal) systems, this goal is less relevant. My POV proposes that cognitive friction is a vital safety feature, and outlines testable hypotheses and methods to investigate these core questions.

Introduction: The Human Factors Perspective

As a Researcher with a Ph.D. in Human Factors, my approach to designing interfaces for autonomous systems diverges from traditional consumer tech philosophies and heuristics. In environments like Google or Apple, the ultimate goal of UX is to remove friction. For example, a common mantra is that "the interface should disappear," allowing the user to accomplish their task with minimal cognitive demand. However, when we apply this "frictionless" philosophy to autonomous weapons platforms and command-and-control software, we encounter a severe ethical and operational hazard.

If we make a system too easy to use, we invite *automation bias* and *complacency* (two well-studied constructs in Human Factors research). Operators may over time cease to function as analytical decision-makers and instead become passive approvers of machine logic and decision-making. To address this risk, my point of view is that **we must design for calibrated cognitive friction**. We must intentionally insert moments of required analytical thought and cognitive effort into the UI to ensure the human is not merely *in* the loop, but actively *controlling* the loop.

Question 1: The Responsibility Gap

THE QUESTION

"As autonomous systems are deployed faster than the frameworks that govern them, who is responsible for keeping humans meaningfully in the loop?"

In the absence of comprehensive and up-to-date Department of Defense (DoD) frameworks, the responsibility for Meaningful Human Control defaults to the system creators. Defense companies cannot rely on policy to define interface safety when the deployment speed outpaces the speed of oversight and bureaucracy. The burden falls on the engineers, designers, and researchers at defense tech organizations to embed governance directly into the software stack.

HYPOTHESIS 1 (RESPONSIBILITY ATTRIBUTION)

Operators implicitly attribute the responsibility for maintaining safety and legal compliance to the system designers. Consequently, if an interface permits an action, operators will assume that action falls within authorized operational and ethical bounds, reducing their own perceived need for independent verification.

Proposed Methodology

To test this, we will conduct a **Vignette-Based Cognitive Walkthrough Study** combined with a modified *Moral Foundations Questionnaire*.

- **Participants:** 30 active-duty or recently separated military operators familiar with Command and Control systems.
- **Procedure:** Participants will be presented with high-fidelity mockups displaying ambiguous threat scenarios. Half of the participants (Group A) will use an interface with strict, system-enforced safety checklists. The other half (Group B) will use an open-ended interface where the system provides a recommendation but leaves the final verification entirely to the user's discretion.
- **Metrics:** We will measure the *Locus of Accountability* using post-scenario interviews and Likert-scale surveys. We will ask participants who they believe is at fault if the scenario results in an erroneous engagement: the operator, the software, or the chain of command.
- **Expected Outcome:** We expect Group B to experience higher anxiety but take more time to verify data, while Group A will defer to the system, proving that UX constraints fundamentally shift where the operator believes responsibility lies.

Question 2: The Friction Paradox

THE QUESTION

"Does better UX on autonomous systems actually reduce harm, or does it make harm easier to authorize?"

This question highlights a dangerous trap in defense UX. "Better" UX is traditionally defined as faster, easy to use, and requiring fewer clicks. In the context of defense tech systems, making harm "easier to authorize" is a significant failure condition. For example, if an interface makes it effortless for an operator to agree with an AI's potential attack recommendation, this decision will bypass the slower, more thoughtful reflection of the operator's prefrontal cortex that is responsible for complex reasoning and moral judgment.

HYPOTHESIS 2 (CALIBRATED FRICTION)

Introducing forced "micro-frictions" will increase task completion time by 15-20% but will decrease the rate of false-positive authorizations by over 50%.

Proposed Methodology

Execute an **A/B Simulator Experiment** utilizing **Dual-Task Paradigms**.

- **Setup:** Participants will be placed in a simulated command center environment under moderate time pressure. They will monitor a sector with mixed civilian and adversary drone traffic.
- **Variant A ("Seamless"):** The system auto-flags targets. The operator simply clicks "Approve" or "Reject."
- **Variant B ("Friction Gate"):** The system auto-flags targets, but to approve, the operator must interact with a "friction gate." For example, they must highlight the specific sensor data point (e.g., thermal signature, speed) that justifies the engagement before the "Approve" button unlocks.
- **Metrics:** Response Time (RT), Accuracy (Hit rate, False Alarm rate, Miss rate), and Eye-tracking data to measure fixation duration on critical information versus the "Approve" button.
- **Expected Outcome:** Variant A will result in faster responses but a higher rate of engaging friendly/neutral targets (automation bias). Variant B will show longer fixation times on data and significantly higher accuracy, proving that "harder to use" can mean "safer to operate."

Question 3: The "Moral Crumple Zone" at Scale

THE QUESTION

"If the human is accountable, but the system is making recommendations at a speed and scale that no human can meaningfully evaluate - are they really in the loop, or is this just a cover for an automated decision?"

This question is related to the concept of the "Moral Crumple Zone" (M.C. Elish, 2019). Similar to how the crumple zone in a car is designed to absorb a crash, the operator in a highly automated system is often positioned to absorb the moral/legal liability of a system failure. For example, when a system presents 50 target recommendations a minute, the operator cannot evaluate the underlying data. They are forced to rely on heuristics (e.g., "the system is usually right, so I will approve"). At this scale, the operator is not a true decision-maker.

HYPOTHESIS 3 (COGNITIVE OVERLOAD THRESHOLD)

There is a measurable threshold of system recommendations per minute (RPM) beyond which human operators abandon analytical verification and default to pure compliance. Beyond this threshold, Meaningful Human Control ceases to exist, regardless of the user interface design.

Proposed Methodology

- **Procedure:** Operators will monitor an autonomous system making recommendations. We will incrementally increase the rate of recommendations from 5 RPM to 60 RPM over a 45-minute session. Interspersed within these recommendations will be deliberate system errors (false positives).
- **Metrics:** 1. *Behavioral:* The rate at which operators catch deliberate system errors. 2. *Subjective:* NASA Task Load Index (NASA-TLX). 3. *Physiological:* Heart Rate Variability (HRV) and Galvanic Skin Response (GSR).
- **Expected Outcome:** We will identify an inflection point (the "Accountability Horizon"). Before this point, operators reliably catch errors and show engaged physiological markers. After this point, error detection will plummet, and HRV will indicate distress.

Conclusion and Next Steps

The transition from consumer product usability to defense-grade, human-machine teaming requires a shift in design philosophy. We cannot import UX principles that were designed to keep users scrolling on social media or buying products with one click. In the context of autonomous platforms, UX must demand attention, enforce accountability, and align with the limits of human cognition.

I believe the research methods outlined above will provide empirical data to answer the three questions posed at the Learners conference. By testing the Locus of Accountability, quantifying the Friction Paradox, and defining the cognitive limits of the "Loop," we can develop a framework for **Human-Centric Defense Design**. This will not only make these automated systems safer and more effective but will position the organization as a thought leader in the ethical deployment of artificial intelligence in modern warfare.